

UNIT-3

Chapter -5

5. Power plants

A power plant is an industrial facility used to generate electric power with the help of one or more generators which converts different energy sources into electric power.

Electricity is produced at an electric power plant. Some fuel source, such as coal, oil, natural gas, or nuclear energy produces heat. The heat is used to boil water to create steam. The steam under high pressure is used to spin a turbine.

For this reason, a power generating station has to not only take care of efficient generation but also the fact that the power is transmitted efficiently over the entire distance and that's why, the transformer switch yard to regulate transmission voltage also becomes an integral part of the power plant.

5.1 Types of Power Plants: A power plant can be of several types depending mainly on the type of fuel used. A power generating station can be broadly classified in to 5 types mentioned below.

1. Thermal Power Plants
2. Diesel Engine Power Plants
3. Gas Turbine Power Plants
4. Nuclear Power Plants
5. Hydro Electric Power Plants

5.1.1 Introduction to Thermal Power Station:

A thermal power station or a coal fired thermal power plant is the most conventional method of generating electric power with reasonably high efficiency. The layout of a diesel power plant is given in Fig. 5.1. It uses coal as the primary fuel to boil the water available to superheated steam for driving the steam turbine. The steam turbine is then mechanically coupled to an alternator rotor, the rotation of which results in the generation of electric power. Generally in India, bituminous coal or brown coal are used as fuel of boiler which has volatile content ranging from 8 to 33% and ash content 5 to 16 %. To enhance the thermal efficiency of the plant, the coal is used in the boiler in its pulverized form.

In coal fired thermal power plant, steam is obtained in very high pressure inside the steam boiler by burning the pulverized coal. This steam is then super heated in the super heater to extreme high temperature. This super heated steam is then allowed to enter into the turbine, as the turbine blades are rotated by the pressure of the steam. The turbine is mechanically coupled with alternator in a way that its rotor will rotate with the rotation of turbine blades. After entering into the turbine, the steam pressure suddenly falls leading to corresponding increase in the steam

volume. After having imparted energy into the turbine rotors, the steam is made to pass out of the turbine blades into the steam condenser of turbine. In the condenser, cold water at ambient temperature is circulated with the help of pump which leads to the condensation of the low pressure wet steam.

In thermal power plants, the heat energy obtained from combustion of solid fuel (mostly coal) is used to convert water into steam; this steam is at high pressure and temperature. This steam is used to rotate the turbine blade turbine shaft is connected to the generator.

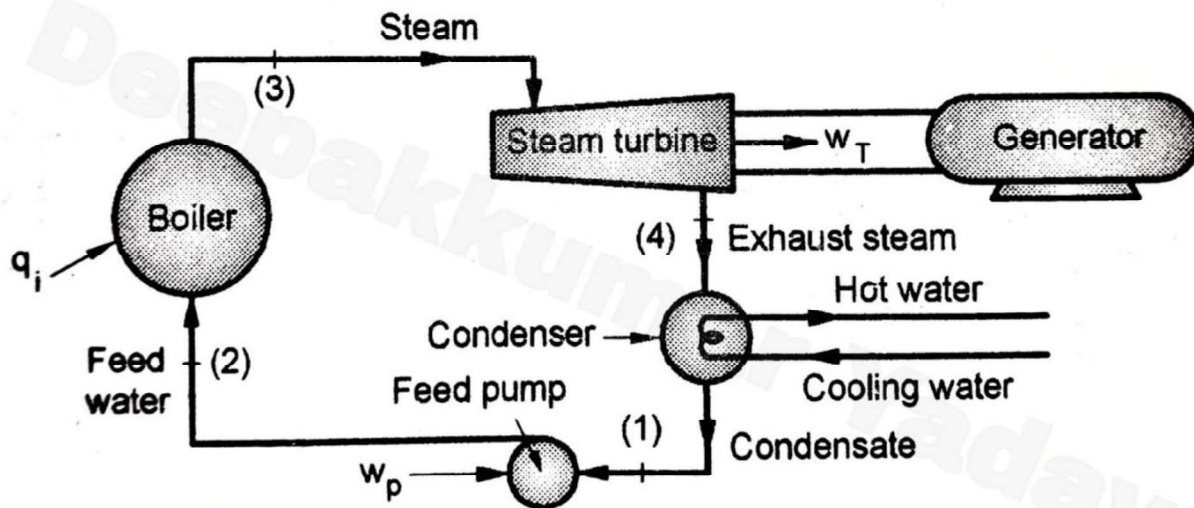


Fig 5.1 Layout of Thermal Power Plant

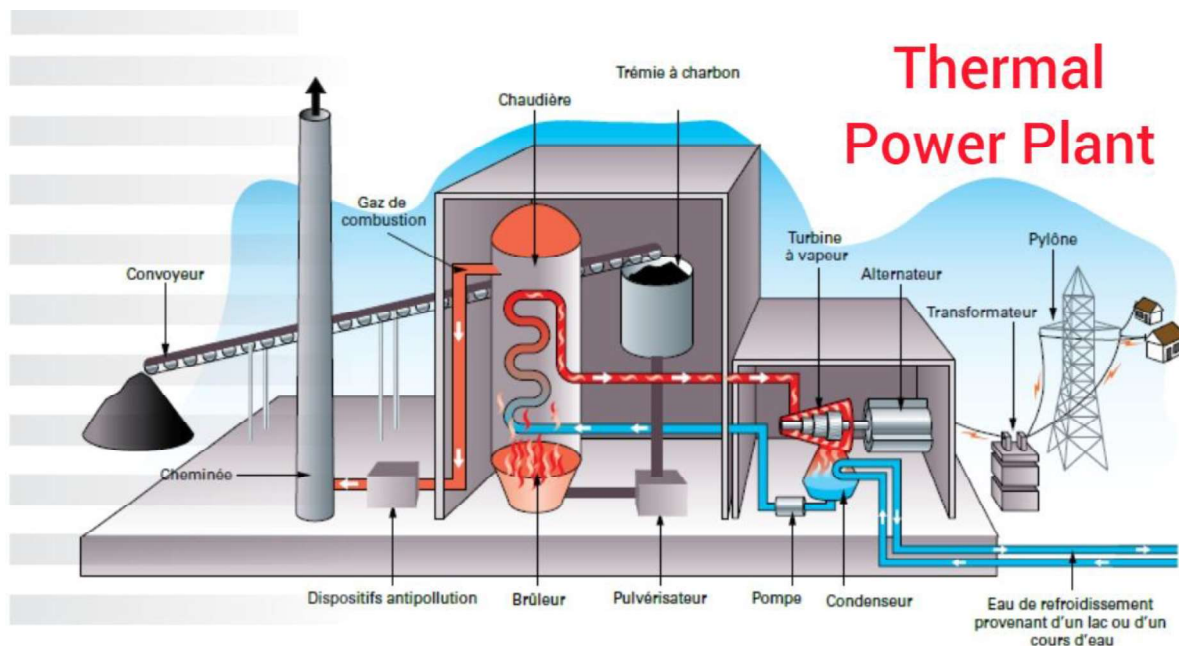


Fig 5.2 Thermal Power Plant

5.1.2 Introduction to Diesel Engine Power Plants:

The layout of a diesel power plant is given in Fig. 8.6. Normally, multi cylinder 2-stroke turbocharged diesel engines are used in power plants (for the principle of working of a diesel engine, please refer to chapter on IC Engine). In a turbocharged engine, the atmospheric air is compressed by a compressor run by an exhaust-driven gas turbine and the compressed air is taken inside the cylinder. Due to this, mass of air intake and amount of fuel burnt will be considerably increased giving rise to increased output power and higher thermal efficiency. Due to turbo charging, the operating temperature of the engine is increased. So, the lubricating oil coming out of the engine should be cooled in oil cooler. The cooling water from the engines is normally cooled in a spray tank and recirculated. Due to high capacity, the engine is started by using compressed air.

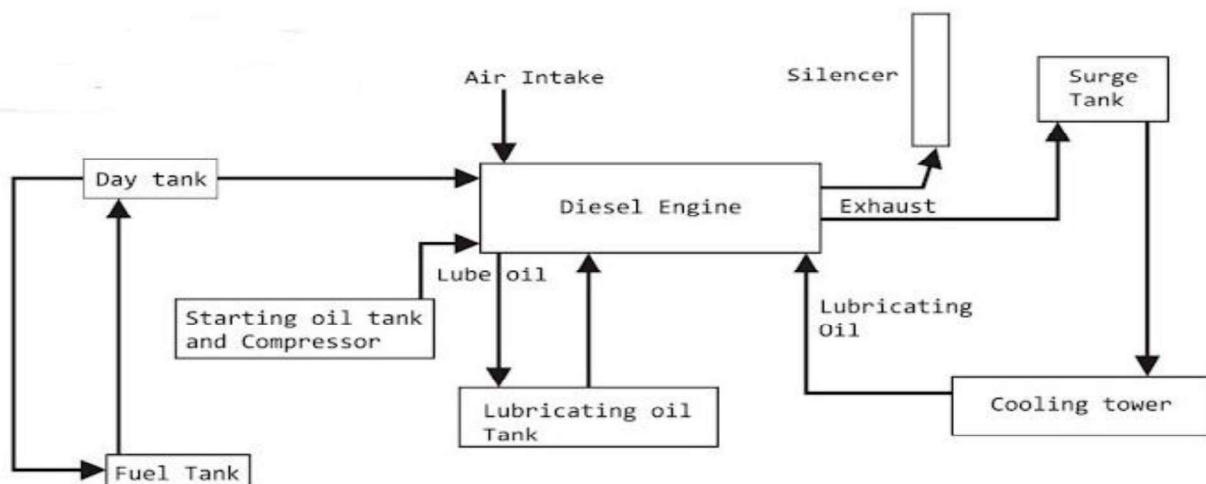


Fig 5.3 Layout of Diesel Engine Power Plant

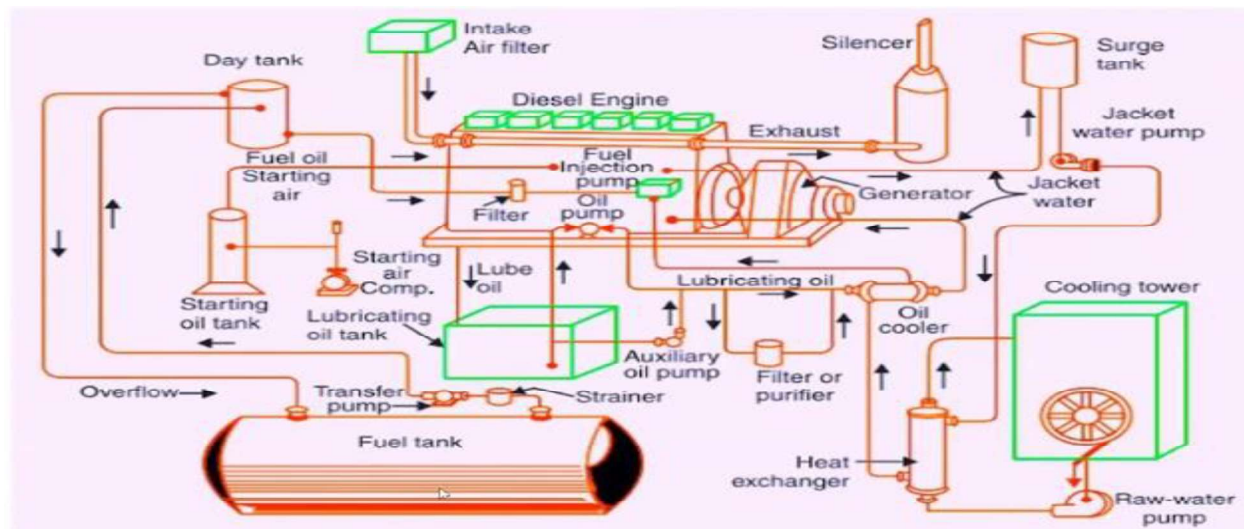


Fig 5.4 Diesel Engine Power Plant

5.1.2.1 Advantages of diesel Power Plant:

1. Plant layout is simple.
2. Easy to install and commission the plant, in comparison with the steam power plant or hydro-plant.
3. Quick starting and easy pick-up of loads are possible in a very short time.
4. Power plant can be located near the load centre.
5. The load operation is easy and requires minimum technical staff.
6. Efficiency at part loads does not fall low.
7. Fuel handling is easier and no problem of ash disposal exists.

5.1.2.2 Disadvantages of diesel Power Plant

1. Maximum capacity of the plant is limited to about 50 MW of power.
2. Diesel fuel is much more expensive than coal.
3. Cost of maintenance and lubrication is high.
4. Overload running may not be possible.

5.1.3 Introduction to Hydro Electric Power Plants:

In a hydroelectric power plant, the potential energy of water stored in a dam is made use of in running a water turbine coupled to an electrical generator. It is estimated that about 23 per cent of the total electric power in the world comes from hydro power. In Tamil Nadu, the total generation of power from hydroelectric plants amounts to 1950 MW and in an all-India level, it amounts to about 18000 MW.

Layout of a Hydroelectric Power Plant The layout of a hydroelectric power plant is given in Fig. The water from the dam is brought to the water turbine by a large diameter penstock pipe. The penstock pipe is made of steel or reinforced concrete. It is desirable to eliminate sharp bends in the penstock pipe to avoid the loss of head and special anchoring.

Depending upon the load on the turbine, the amount of water needed is controlled automatically by a valve operated by a centrifugal governor. In case the amount of water is suddenly reduced or stopped by the governor mechanism, water coming down with a high velocity will produce turbulence resulting in a water hammer in the pipe. The penstock pipe may be damaged due to the water hammer. To prevent this, a surge tank is provided. Surge tank is a large closed tank, which will get filled up with water in the event pressure raise in the penstock and the air inside the surge tank shall get compressed. This will safeguard the turbine plant, control valve and the penstock itself. When pressure becomes normal, the accumulated water will flow back to the penstock and to the turbine to produce power. From the turbine, water is allowed to pass through a draft tube to the tail race. The tail race is the water path leading the discharge water from the turbine to the river or canal.

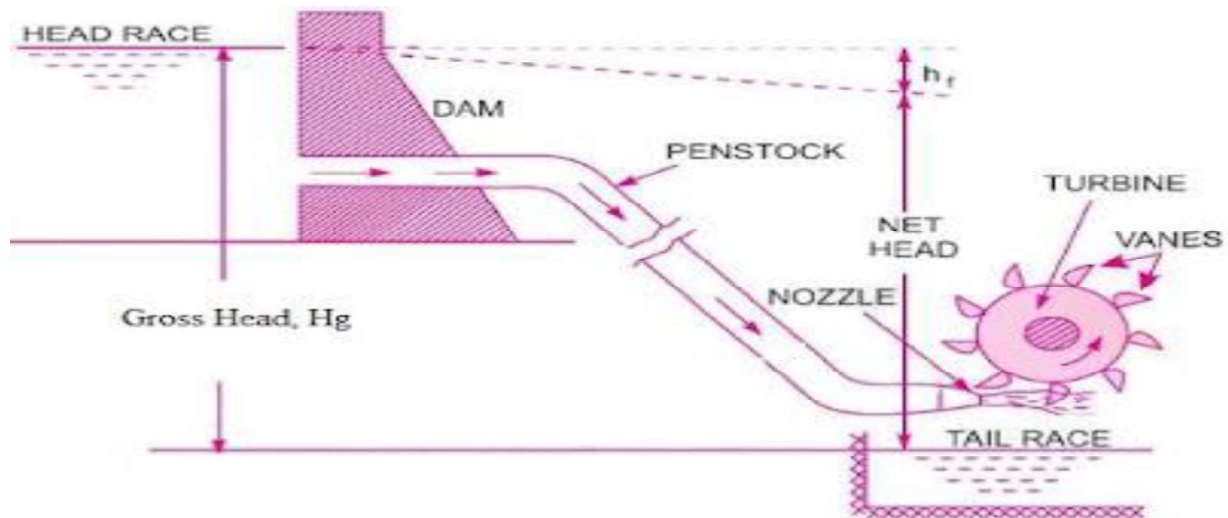


Fig 5.5 Layout of Hydro Electric Power Plants

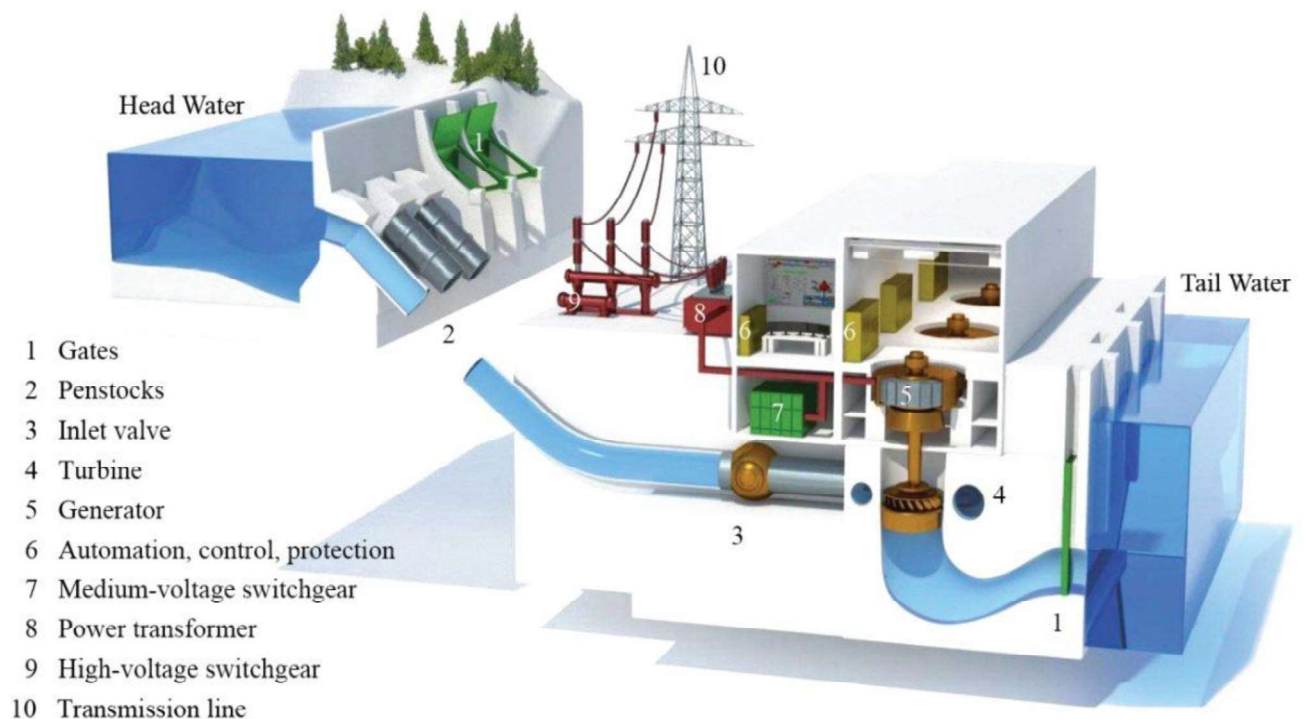


Fig 5.6 Hydro Electric Power Plants

5.1.4 Introduction to Nuclear Power Plants:

Nuclear power plant uses nuclear energy from radioactive element for generating electrical energy. It is generally located far away from populated areas. In future generation of electricity will be depending on Nuclear Power Plant, as it is economical. 1 kg of uranium U - 235 can produce electrical power electrical that can be produced by using 3000 -4500 tonnes of high grade coal or 2000 tonnes of oil.

5.1.4.1 Working Principle of Nuclear Power Plant: The heat generated in the reactor due to the fission of the fuel is taken up by the coolant. The hot coolant then leaves the reactor and flows through the steam generator. In the steam generator the hot coolant transfers its heat to the feed water which gets converted into steam. The steam produced is passed through the turbine, which is coupled with generator. Hence the power is produced during the running of turbine. The exhaust steam from the turbine is condensed in the condenser. The condensate then flows to the steam generator through the feed pump. The cycle is thus repeated.

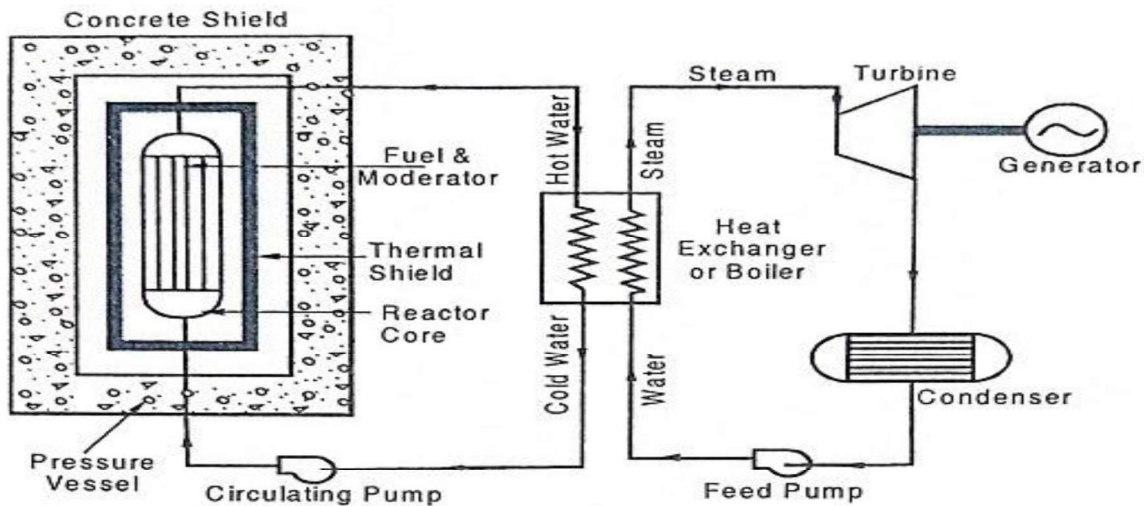


Fig 5.7 Layout of Nuclear Power Plants

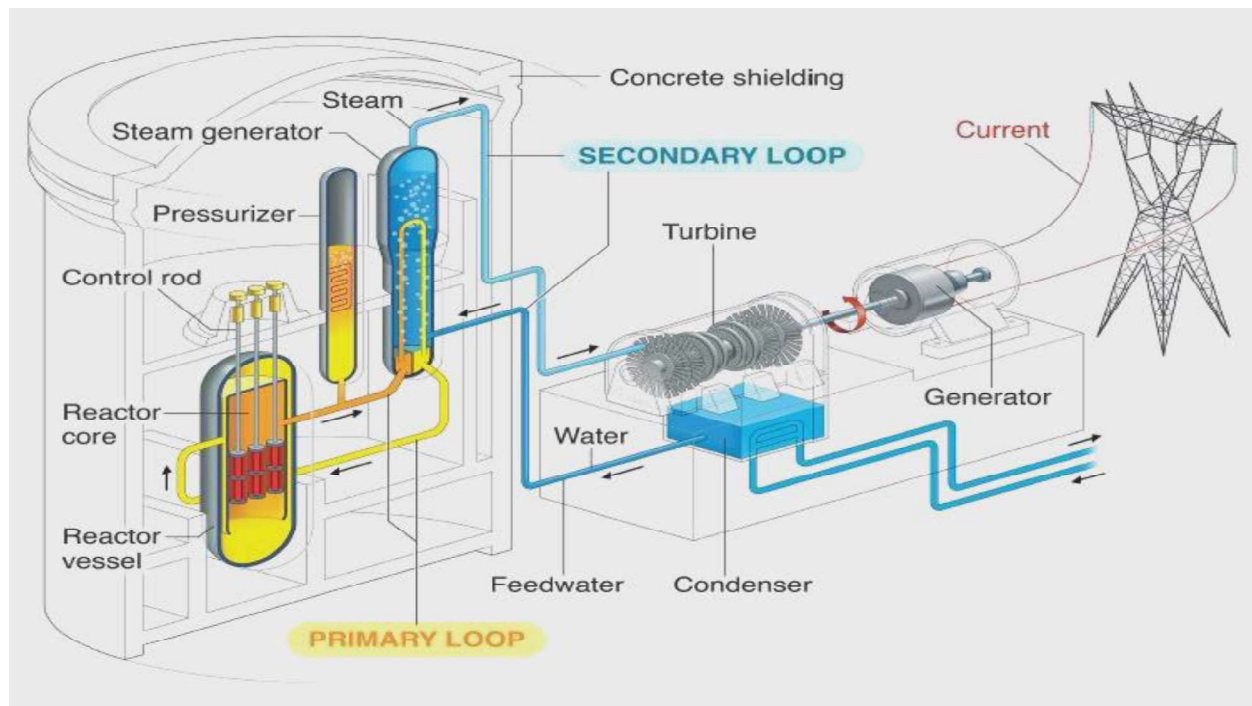


Fig 5.8 Nuclear Power Plants

5.1.4.2 Advantages of Nuclear Power Plant:

1. Requires less space compared to steaming power plant.
2. Fuel required is negligible compared to coal requirement.
3. Fuel transport cost is less. Reliable in operation.
4. Cost of erection is less. Water required is very less.

5.1.4.2 Disadvantages of Nuclear Power Plant:

1. Initial Cost is higher.
2. Not suitable for varying load condition.
3. Radioactive wastes are hazardous.
4. Hence these are to be handled with much care.
5. Maintenance cost is higher.
6. Trained workers are required to operate the plant.

Chapter 6

6. Mechanical Power Transmission

The power is transmitted from one shaft to the other by means of belts, chains and gears. The belts and ropes are flexible members which are used where distance between the two shafts is large. The chains also have flexibility but they are preferred for intermediate distances. The gears are used when the shafts are very close with each other. This type of drive is also called positive drive because there is no slip. If the distance is slightly larger, chain drive can be used for making it a positive drive. Belts and ropes transmit power due to the friction between the belt or rope and the pulley. There is a possibility of slip and creep and that is why, this drive is not a positive drive. A gear train is a combination of gears which are used for transmitting motion from one shaft to another.

6.1 Power Transmission Devices: Power transmission devices are very commonly used to transmit power from one shaft to another. Belts, chains and gears are used for this purpose. When the distance between the shafts is large, belts or ropes are used and for intermediate distance chains can be used. For belt drive distance can be maximum but this should not be more than 10 meter for good results. Gear drive is used for short distances.

6.2 Belts: In case of belts, friction between the belt and pulley is used to transmit power. In practice, there is always some amount of slip between belt and pulleys, therefore, exact velocity ratio cannot be obtained. That is why; belt drive is not a positive drive. Therefore, the belt drive is used where exact velocity ratio is not required.

The following types of belts shown in Figure 3.1 are most commonly used:

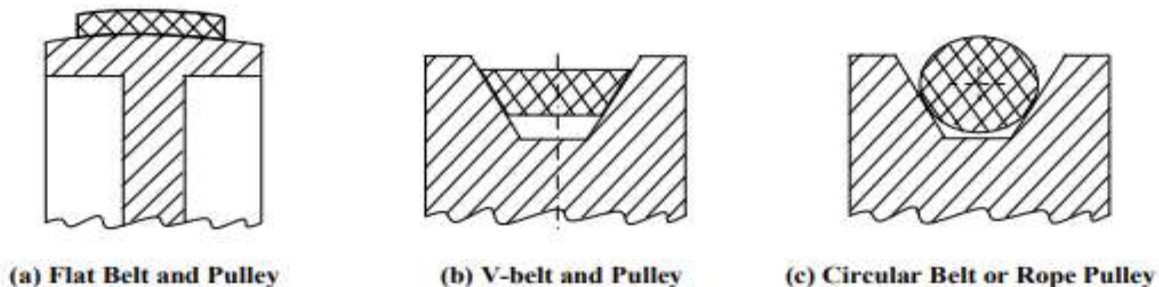


Fig 6.1 Types of belts

The flat belt is rectangular in cross-section as shown in Figure 6.1(a). The pulley for this belt is slightly crowned to prevent slip of the belt to one side. It utilises the friction between the flat surface of the belt and pulley. The V-belt is trapezoidal in section as shown in Figure 6.1(b). It utilizes the force of friction between the inclined sides of the belt and pulley. They are preferred when distance is comparative shorter. Several V-belts can also be used together if

power transmitted is more. The circular belt or rope is circular in section as shown in Figure 6.1(c). Several ropes also can be used together to transmit more power.

The belt drives are of the following types:

- (a) Open belt drive, and (b) Cross belt drive.

a) Open belt drive: Open belt drive is used when sense of rotation of both the pulleys is same. It is desirable to keep the tight side of the belt on the lower side and slack side at the top to increase the angle of contact on the pulleys. This type of drive is shown in fig (a).

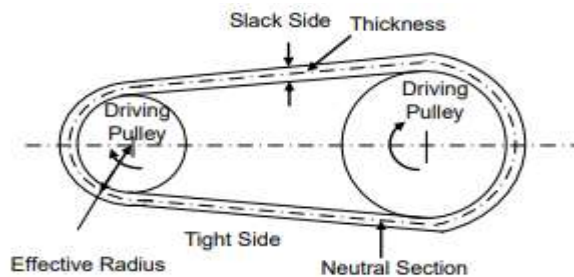


Fig 6.2 (a) Open belt

b) Cross belt drive: In case of cross belt drive, the pulleys rotate in the opposite direction. The angle of contact of belt on both the pulleys is equal. This drive is shown in fig (b). As shown in the figure, the belt has to bend in two different planes. As a result of this, belt wears very fast and therefore, this type of drive is not preferred for power transmission. This can be used for transmission of speed at low power.

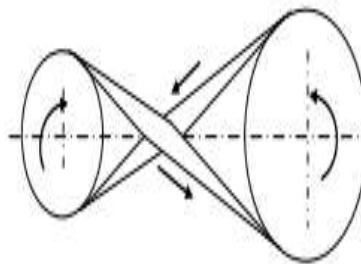


Fig 6.2 (b) Cross belt

6.2.1 Advantages of Belt-drive

1. It can be easily installed and easily removed.
2. The price of the belt drive is low.
3. Simple in construction.
4. The maintenance cost is low.
5. No additional lubricant is required.
6. It can transfer power vertical, horizontal and inclined too.

6.2.2 Disadvantages of Belt-drive

1. Power loss due to slip and creep.
2. Not useful for a very short distance.
3. Speed is limited to some extent.
4. It is quieter.
5. No possibility of a longer life.

6.2.2 Applications of Belt-drive

1. The belt drive is used in the Mill industry.
2. The belt drive is used in Conveyor.
3. Power transmission in industrial machinery and equipment
4. Automotive applications, in different types of engines and transmissions
5. Agricultural machinery, such as tractors and harvesters
6. Conveyor systems in manufacturing and distribution facilities
7. Office equipment, such as printers and copiers
8. Power tools, such as table saws and drill presses.

6.3 Chain drive:

The belt drive is not a positive drive because of creep and slip. The chain drive is a positive drive. Like belts, chains can be used for larger centre distances. They are made of metal and due to this chain is heavier than the belt but they are flexible like belts. It also requires lubrication from time to time. The lubricant prevents chain from rusting and reduces wear. The chain and chain drive are shown in Figure 3.4. The sprockets are used in place of pulleys. The projected teeth of sprockets fit in the recesses of the chain. The distance between roller centers of two adjacent links is known as pitch. The circle passing through the pitch centers is called pitch circle.

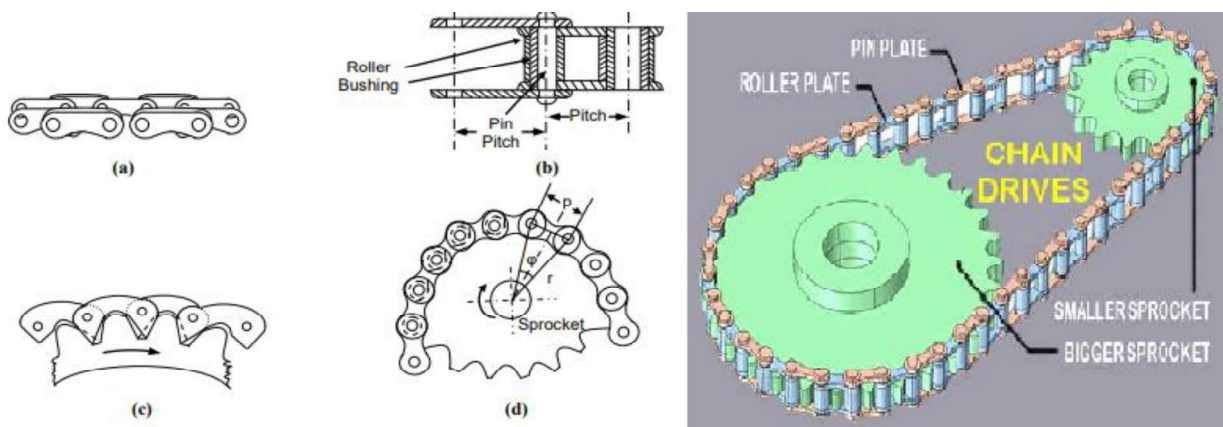


Fig 6.3 Chain drive

6.3.1 Advantages of Chain Drive:

1. They can be used for both long and short distances
2. Several shafts can be driven from a single chain
3. They are compact and have small overall dimensions
4. They have very high efficiency (up to 96%)
5. No slip and creep during the power transmission, ensure the perfect velocity ratio.
6. Chain drives have a low maintenance cost

6.3.2 Disadvantages of Chain drive:

1. Chain drive requires frequent lubrication
2. Load capacity less compared with gear drives
3. Conventional chain drive suffers vibrations due to the chordal effect.
4. They are not suitable for non-parallel shafts

6.3.3 Application of chain drive:

1. Hydraulic lift truck fork operation
2. Increasing or decreasing a driver's output speed by altering gear ratios between the driver and the sprocket being driven
3. Overhead hoists
4. Operating conveyor belts
5. Chain drive used in several industries for several purposes like Transportation Industry, Agriculture Machinery, Material Handling Equipment, Building construction.

6.4 Gears: Gears are also used for power transmission. This is accomplished by the successive engagement of teeth. The two gears transmit motion by the direct contact like chain drive. Gears also provide positive drive. The drive between the two gears can be represented by using plain cylinders or discs 1 and 2 having diameters equal to their pitch circles as shown in Figure 3.5. The point of contact of the two pitch surfaces shall have velocity along the common tangent. Because there is no slip, definite motion of gear 1 can be transmitted to gear 2 or vice-versa.

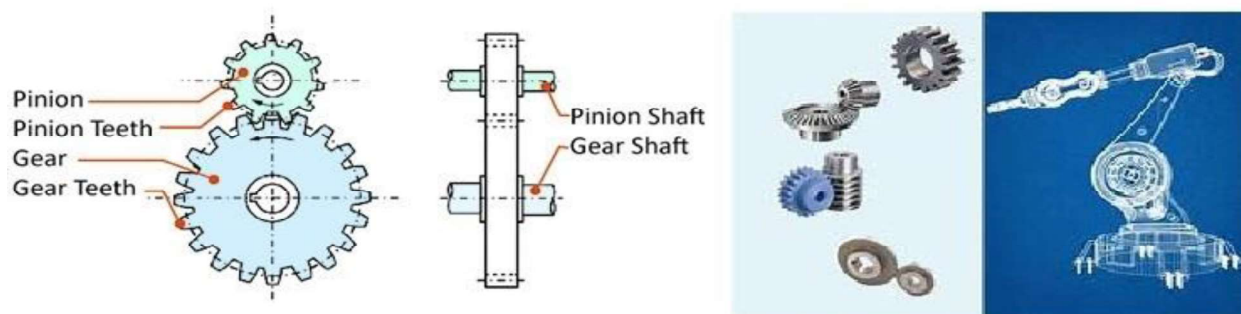


Fig 6.4 Gear drive

6.4.1 Advantages of Gears:

1. Gears are mechanically strong, they can carry higher loads easily.
2. They are used to transmit motion over shafts with small center distances.
3. These mechanical devices are capable of transmitting large high frequency (H.F.).
4. Gears only need to be greased, therefore maintenance is minimal.
5. Gear systems allow us to transfer motion across non-parallel intersecting shafts.
6. Because of its long lifespan, the gear system is extremely compact.

6.4.2 Disadvantages of Gears:

1. Gear systems are not suitable for high-speed operation.
2. They are poor at transmitting motion over long distances.
3. Some noise may come from the operating gear.
4. The engagement of toothed gear wheels under heavy loading might cause irreversible damage to some machine elements.

6.4.3 Application

1. Hydraulic lift truck fork operation
2. Increasing or decreasing a driver's output speed by altering gear ratios between the drivers
3. Overhead hoists
4. Operating conveyer belts
5. Used in several industries for several purposes like Transportation Industry, Agriculture Machinery, Material Handling Equipment, Building construction.

Chapter 7

Introduction to Robotics

The field of robotics has its origins in science fiction. The term robot was derived from the English translation of a fantasy play written in Czechoslovakia around 1920. It took another 40 years before the modern technology of industrial robotics began. Today, Robots are highly automated mechanical manipulators controlled by computers. In this chapter, We survey some of the science fiction stories about robots, and we trace the historical development of robotics technology.

7.1 Robotics: Robotics is an applied engineering science that has been referred to as a combination of machine tool technology and computer science. It includes machine design, production theory, micro electronics, computer programming & artificial intelligence.

OR

"Robotics" is defined as the science of designing and building Robots which are suitable for real life application in automated manufacturing and other non-manufacturing environments.

7.1.2 Industrial robot: The official definition of an industrial robot is provided by the robotics industries association (RIA). Industrial robot is defined as an automatic, freely programmed, servo-controlled, multi-purpose manipulator to handle various operations of an industry with variable programmed motions.

7.2 Components of the Industrial Robotics:

Main components of robot: A typical stand-alone robot shown in fig 7.1 below comprises of the following basic components, namely.

1. Manipulator
2. Sensors devices
3. Robot Tooling
4. Robot controller unit (RCU)

1. Manipulator: It consists of base, arm, & wrist similar to a human arm. It also includes power source either electric, hydraulic or pneumatic on receiving signals from robot controller this mechanical unit will be activated. The movement of manipulator can be in relation to its coordinate system. Which may be Cartesian, cylindrical etc, Depending on the controller, movement may be point to point motion or continuous motion.

The manipulator is composed of 3 divisions,

- i. The major linkages
- ii. The minor linkages (wrist components)
- iii. The end effectors (gripper or tool)

2. Sensors devices: These elements inform the robot controller about the status of the manipulator. These sensors can be either analog or digital and combination. These are

i. visual

ii. Non visual

3. Robot Tooling: Robot tooling is nothing but hand or gripper of the robot also called as “End effectors”. It is provided at the end of the arm. Its design depends on the nature of the work to be performed by the robot.

4. Robot controller unit (RCU): The instructions to the robot to perform the desired tasks are input through the key board of this unit. The controller converts the input programs to suitable signals which activate the manipulator to perform the desired tasks.

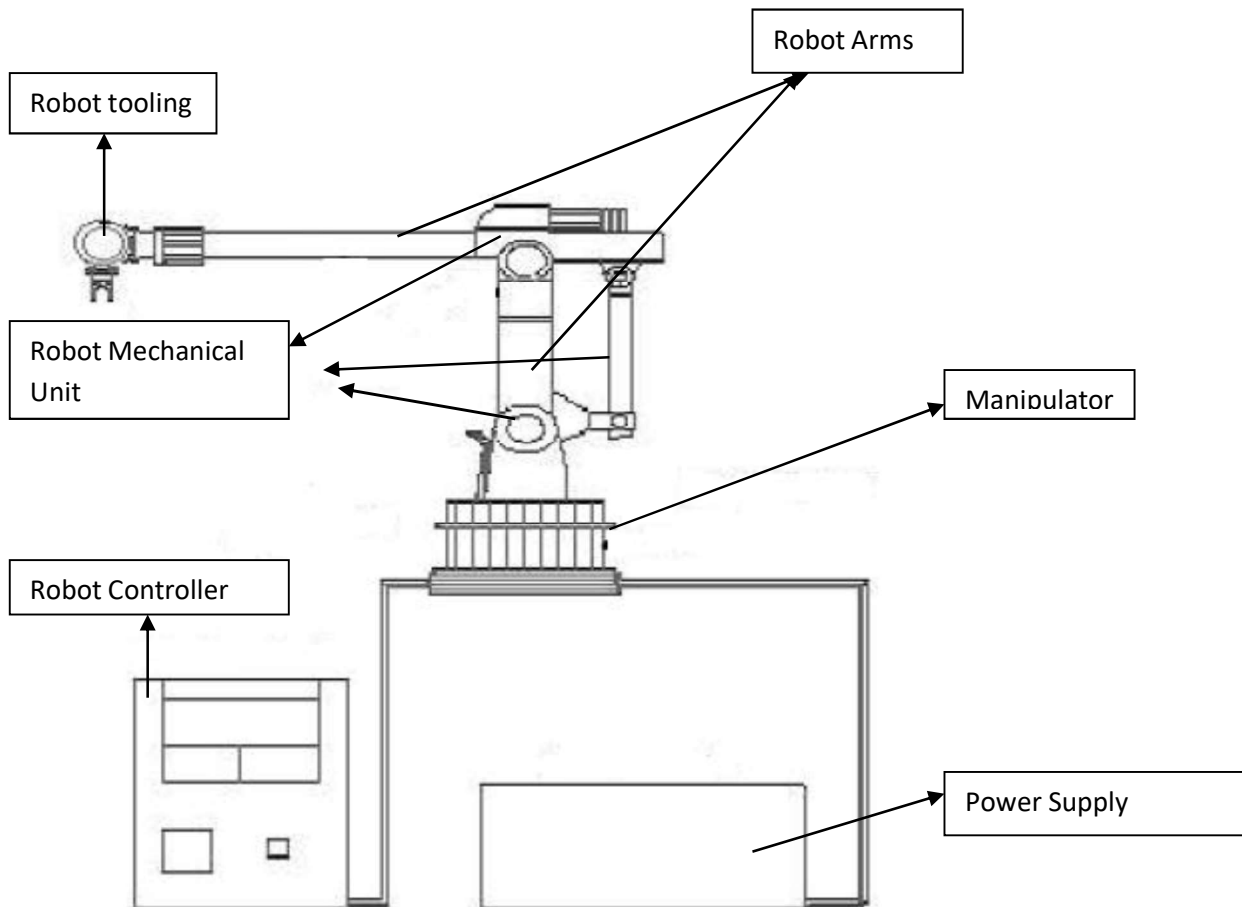


Fig 7.1 Components of the Industrial Robotics

7.3 Classification of Robots (or) Classification by co-ordinate system and control system:

A) Co-ordinate systems: Industrial robots are available in a wide variety of sizes, shapes, and physical configurations. The majority of Today's commercially available robots possess one of the basic configurations:

1. Polar configuration
2. Cylindrical configuration
3. Cartesian coordinate configurable
4. Jointed-arm configuration

1. Polar configuration:- The polar configuration is pictured of Fig 7.2. It uses a telescoping arm that can be raised or lowered about a horizontal pivot the pivot is mounted on a base. These various joints provide the robot with the capability to move its arm within a spherical space, and hence the name “Spherical Coordinate” robot is some times applied to this type. A number of commercial possess the polar configuration.

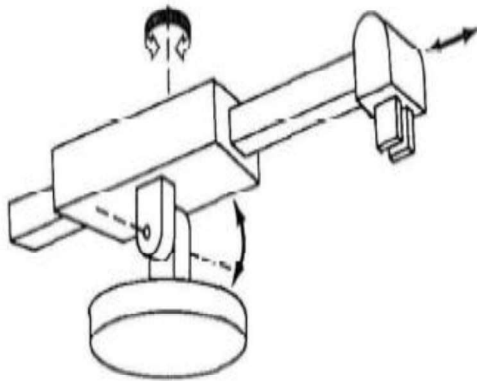


Fig 7.2 Spherical Coordinate

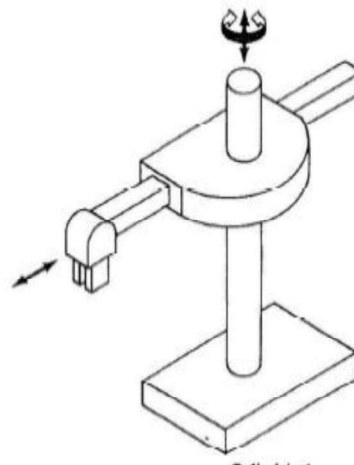


Fig 7.3 Cylindrical Coordinate

2. Cylindrical configuration:- The cylindrical configurable, as shown in fig 7.3, uses a vertical column and a slide that can be moved up or down along the column. The robot arm is attached to the slide so that it can be moved radially with respect to the column. By rotating the column, the robot is capable of achieving a work space that approximates a cylinder.

3. Cartesian coordinate configurable:

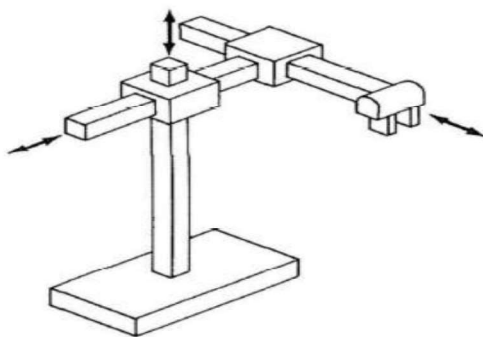


Fig 7.4 Spherical Coordinate

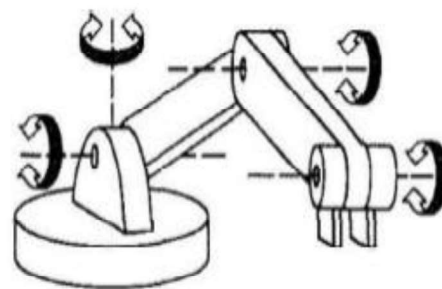


Fig 7.5 Cylindrical Coordinate

The Cartesian coordinate robot of Fig 7.4 uses three perpendicular slides to construct the x, y, and z axes. Other names are sometimes applied to this configuration, including xyz robot and rectilinear robot. By moving the three slides relative to one another, the robot is capable of operating within a rectangular work envelope.

4. Jointed-arm configuration: The jointed-arm robot is pictured in Fig 7.5. Its configuration is similar to that of the human arm. It consists of two straight components. Corresponding to the human forearm and upper arm, mounted on a vertical pedestal. These components are connected by two rotary joints corresponding to the shoulder and elbow.

7.4 Different types of joints are used in robots:-

The different types of joint are used in robots are,

1. Prismatic Joint (Linear Motion)
2. Revolute Joint (Rotational Motion)
3. Screw Joint (Linear and Rotational Motion)
4. Spherical Joint
5. Planar Joint
6. Cylindrical Joint
7. Twisting Joint

7.4.1. Prismatic Joint (Linear Motion): Prismatic joints are also called sliders, constituting purely linear motion along the joint axis. The joint slides in one, of the two directions along a singular axis. This type of motion is common in hydraulic or pneumatic cylinders, there is no rotation. The degree of freedom is one (i.e., translation).

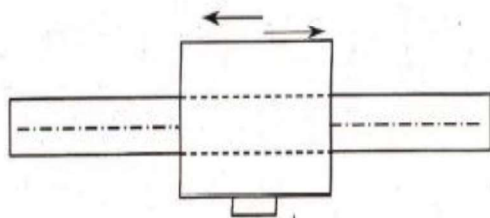


Fig 7.6 Prismatic Joint

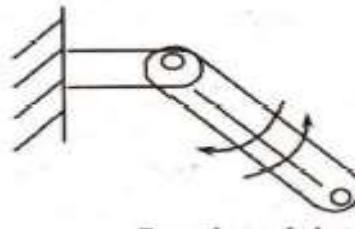


Fig 7.7 Cylindrical Coordinate

7.4.2. Revolute Joint (Rotational Motion): Revolute joints constitute purely rotation motion along the joint axis. Revolute joints are most commonly found in industrial robots. The degree of freedom is one. There is no translation motion.

Examples: i. Elbow joint, waist joint. ii. Most revolute joints cannot rotate through a full 360° , but are mechanically constrained.

7.4.3. Screw Joint (Linear and Rotational Motion): Screw joints are a combination of first two types of joints. They constitute a simultaneous rotation and linear motion along a joint axis. Screw joints are more often used in tools for a robot end effector rather than a joint of motion for a robot. This joint follows the thread of the axis in spiral to move along axis.

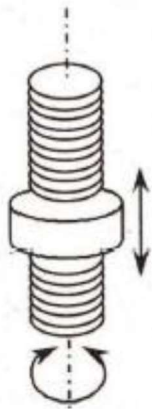


Fig 7.8 Screw Joint

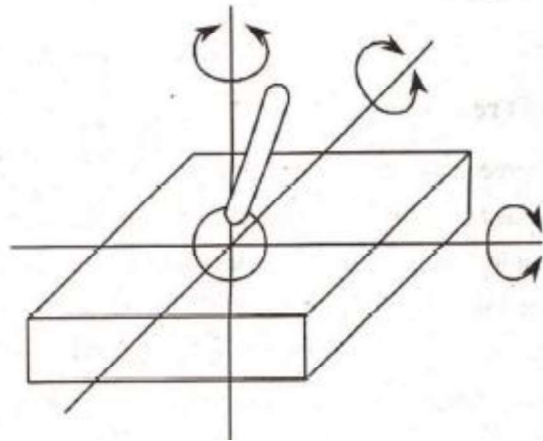


Fig 7.9 Spherical Joint

7.4.4. Spherical Joint: Spherical joints are most utilized joints and just slide causing a revolving movement.

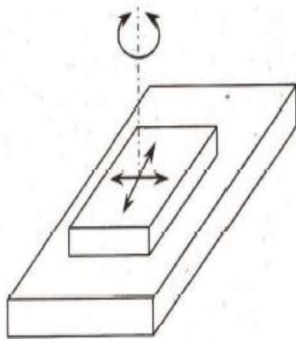


Fig 7.9 Planar Joint



Fig 7.10 Cylindrical Joint

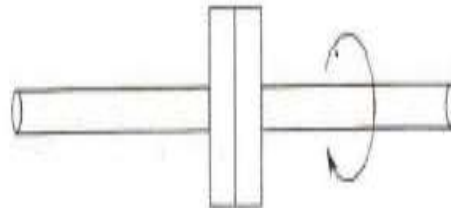


Fig 7.11 Twisting Joint

7.4.5. Planar Joint: It consists of three degrees of freedom as the figure 7.9 i.e., one rotary and two sliding motions.

7.4.6. Cylindrical Joint: It consists of two degrees of freedom as shown in figure 7.10 i.e., one rotary and one sliding motion

7.4.7. Twisting Joint: It consists of single degree of freedom as shown in the figure 7.11.

7.5 Types of Mechanical gripper actuations:- There are various ways of classifying mechanical grippers and their actuating mechanisms. One method is according to the type of

finger movement used by the gripper. In this classification, the grippers can actuate the opening and closing of the Rogers by one of the following motions:

1. Pivoting movement
2. Linear or translational movement

In the pivoting movement. The fingers rotate about fixed pivot points on the gripper to open and close. The motion is usually accomplished by some kind of linkage mechanism. In the linear movement, the Hagen open and close by moving in parallel to each other. This is accomplished by means of guide rails so that each finger base slides along a guide rail during actuation. The translational finger movement might also be accomplished by means of a linkage which would maintain the Rogers in a parallel orientation to each other during actuation. Mechanical grippers can also be classed according to the type of kinematic device used to actuate the Roger movement.

In this classification we have the following types:

- i. Linkage actuation,
- ii. Gear & rack actuation,
- iii. Cam actuation,
- iv. Screw actuation,
- v. Rope & pulley actuation.

- i. **Linkage actuation:** The linkage category covers a wide range of design possibilities to actuate the opening and closing of the gripper. The design of the linkage determines how the input force to the gripper is converted into the gripping force applied by the fingers. The 'linkage configuration also determines other operational features such as how wide the gripper fingers will open and how quickly the gripper will actuate.

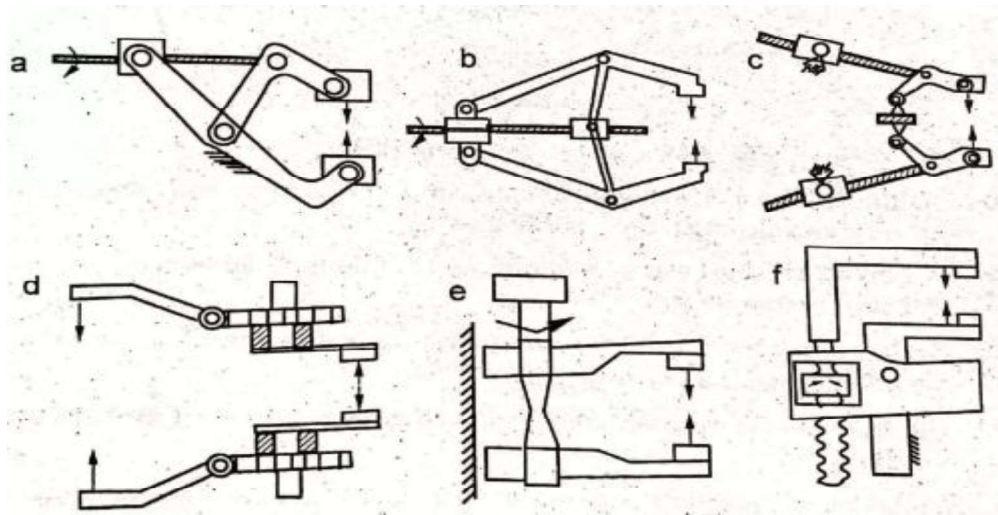


Fig 7.12 Linkage actuation gripper mechanism

- ii. **Gear & rack actuation:** These are some other mechanism that would provide a linear motion, movement of the rack would drive to partial pinion gears and these would in turn open and close the finger.

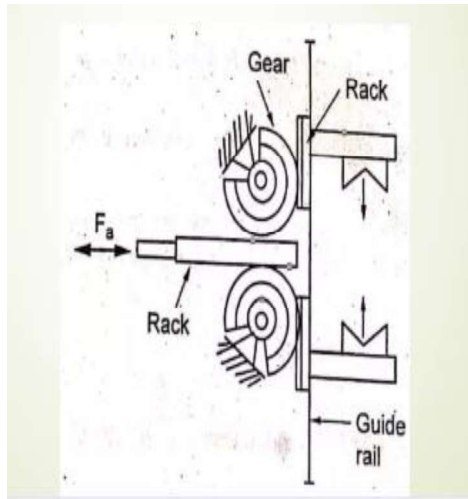


Fig 7.13 Gear & rack actuation

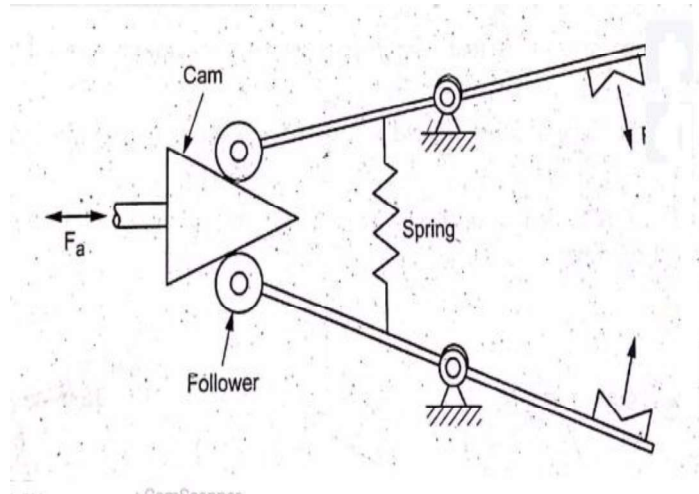


Fig 7.14 Cam actuation:

iii. Cam actuation: The cam actuated gripper Includes a variety possible designs, one of which is shown in Fig 7.14. A cam and follower arrangement, often using a spring-loaded follower, can provide the opening and closing action of the gripper. For example, movement of the cam in one direction would force the gripper to open. While movement of the cam in the opposite direction would cause the spring to force the gripper to close. The advantage of this arrangement is that the spring action would accommodate different sized parts. This might be desirable, for example, in a machining operation where a single gripper is used to handle the raw work part and the finished part. The finished part might be significantly smaller after machining.

iv. Screw actuation: The screw is turned by a motor. Usually accompanied by a speed reduction mechanism. When the screw is rotated in one direction, this causes a threaded block to be translated in one direction. When the screw is rotated in the opposite direction. The threaded block moves in the opposite direction. The threaded block is, in turn, connected to the gripper fingers to cause the corresponding opening and closing action.

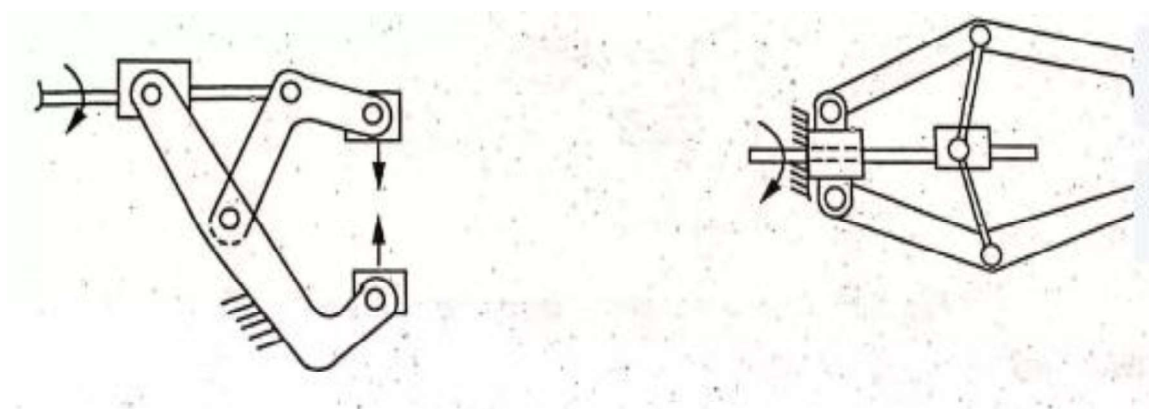


Fig 7.15 Screw actuation

v. Rope & pulley actuation: Rope & pulley actuating can be design to open & closing a mechanical gripper.

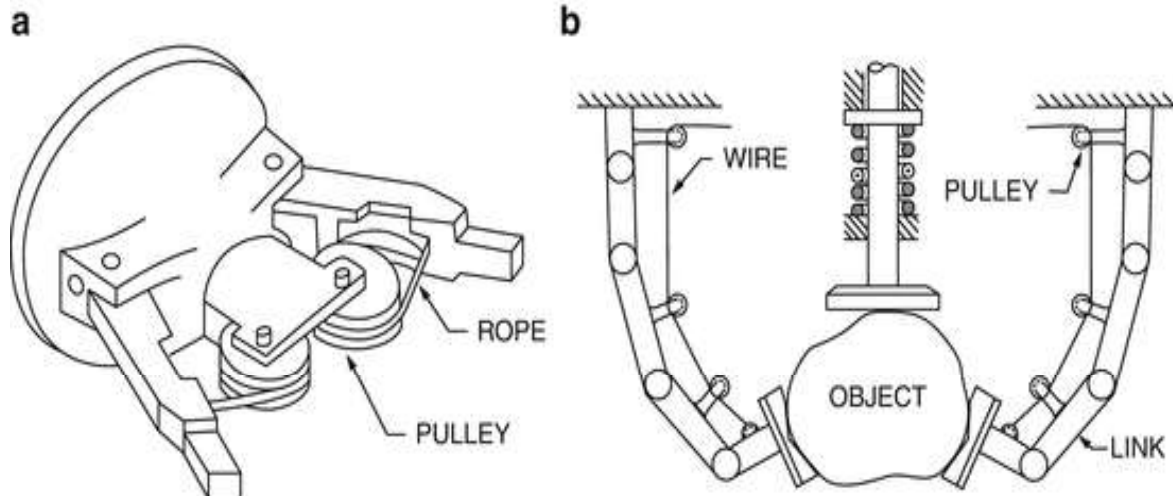


Fig 7.15 Rope & pulley actuation

7.6 OBJECTS USING INDUSTRIAL ROBOT:

1. To increased productivity.
2. To raise the quality level of products.
3. To maintain the labor requirements.
4. To reduce production time.
5. To improve existing manufacturing process.
6. To enhance the life production machines.
7. To increase the safety level of the labor.
8. Thus to minimize the loss of man hours on account of accident and diseases.

7.7 ROBOT CHARACTERISTICS:

1. **Sensing:** First of all your robot would have to be able to sense its surroundings. It would do this in ways that are not similar to the way that you sense your surroundings. Giving your robot sensors like light sensors (eyes), touch and pressure sensors (hands), chemical sensors, hearing sensors, and taste sensors will give your robot awareness of its environment.
2. **Movement:** A robot needs to be able to move around its environment. Whether rolling on wheels, walking on legs or propelling by thrusters a robot needs to be able to move. To count as a robot either the whole robot moves, like the Sojourner or just parts of the robot moves, like the Canada Arm.
3. **Energy:** A robot needs to be able to power itself. A robot might be solar powered, electrically powered, battery powered. The way your robot gets its energy will depend on what your robot needs to do.
4. **Intelligence:** A robot needs some kind of "smarts." This is where programming enters the pictures. A programmer is the person who gives the robot its 'smarts.' The robot will have to have some way to receive the program so that it knows what it is to do.

1.

7.8 ADVANTAGES OF ROBOT:

1. Robots increase productivity, safety, efficiency, quality, and consistency of products.
2. Robots can work in hazardous environments without the need.
3. Robots need no environmental comfort.
4. Robots work continuously without experiencing fatigue of problem.
5. Robots have repeatable precision at all times.
6. Robots can be much more accurate than human.
7. Robots can process multiple stimuli or tasks simultaneously.

7.9 DISADVANTAGES OF ROBOT:

1. High initial cost.
2. Need extra space and new technologies.
3. They need constant power.
4. They lead humans to lose their jobs.
5. Perform relatively few tasks.
6. Need high skilled engineers, programmers.
7. They impacts human interaction
8. Only programmer understands its function

7.10 APPLICATION OF ROBOT:

1. Material handling, Material transfer.
2. Machine loading and/or unloading.
3. Spot welding, Continuous arc welding, Spray coating.
4. Assembly, Inspection.
5. Robots in Space
6. Robots in Hazardous Environments
7. Medical sector
8. Robots at Home (for entertained, cooking and other house hold work)
9. In military service.

7.11 Medical Robotics: Medical robotics is ultimately an application-driven research Field. Development of medical robotic systems requires significant innovation and can lead to very real, fundamental advances in technology, Medical robots must provide measurable and significant advantages if they are to be widely accepted and deployed. These advantages are often difficult to measure, can take an extended period to assess, and may be of varying importance to different groups.

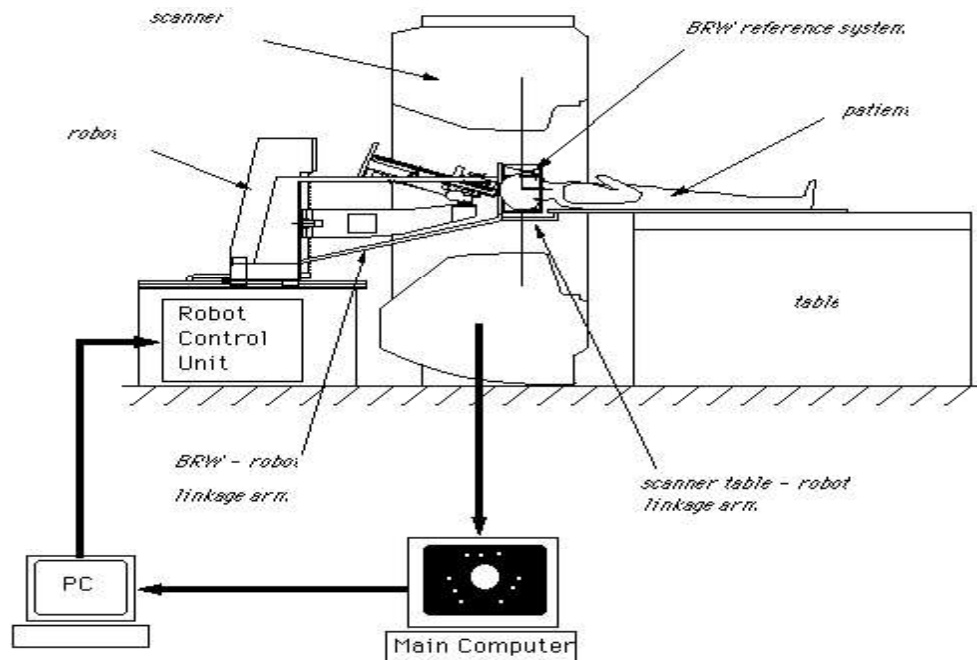


Fig 12: The operating theatre in a robotic surgery

7.10.1 Advantages Medical Robotics:

1. Can significantly improve surgeons' technical capability to perform procedures by exploiting the complementary strengths of humans and robots
2. Medical robots can be constructed to be more precise and geometrically accurate than an unaided human.
3. They can operate in hostile radiological environments and can provide great dexterity for minimally invasive procedures inside the patient's body.
4. These capabilities can both enhance the ability of an average surgeon to perform procedures that only a few exceptionally gifted surgeons can perform unassisted
5. Also makes it possible to perform interventions that would otherwise be completely.

7.10.2 Industrial Applications:

Industrial Robot Applications can be divided into

1. Material handling application :
 - Involve the movement of material or part from one location to another.
 - It includes part placement, palletizing and/ or depalletizing transportation/ conveyance, machine loading and unloading.
2. Processing operation :
 - Requires the robot to manipulate a special process tool as the end effectors.
 - The application include spot welding, arc welding, riveting, spray painting, machining , metal cutting, polishing.
3. Assemble Application :

- Involve part-handling manipulations of a special tools and other automatic tasks and operations.
- 4. Inspection operations :
 - Require the robot to position a work part to an inspection device
 - Involve the robot to manipulate a device or sensor to perform the inspection.

7.10.3 Future and Non-Industrial Robot Applications:

1. House hold applications
 - Cleaning, child's play mate, servant
2. Underwater Applications:
 - Involve prospecting for minerals on the floor of the ocean
3. The medical applications
 - Routine examinations
 - Surgical procedures
4. Surveillance and Guard duty
 - In military , hazardous environments
 - Power generating plants, oil refineries and other facilities.